

Candidate Name

Centre Number

Candidate Number



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Ordinary Level

AGRICULTURE
PAPER 2

4001/2
 2 hours

SPECIMEN PAPER

Additional materials:
 Answer booklet,
 Scientific calculator.

INSTRUCTIONS TO CANDIDATES

Write your name, centre number and candidate number in the spaces provided at the top of this page and on all separate answer booklets used.

Section A

Answer **all** questions.

Write your answers in the spaces provided on the question paper.

Section B

Answer **one** question from this section.

Write your answers in the answer booklet provided.

Section C

Answer **one** question from this section.

Write your answers in the answer booklet provided.

At the end of the examination, fasten the answer booklet used for Section B and C securely to the question paper.

INFORMATION FOR CANDIDATES

The marks are given in brackets [] at the end of each question or part question.

**FOR EXAMINER'S
USE**

Section A	
1	
2	
3	
4	
5	
6	
Section B	
Section C	

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SECTION A [60 marks]

Answer **all** questions in the spaces provided on the question paper.

Candidates are advised to spend approximately **80 minutes** on this section.

- 1 (a) Explain any **two** socio-economic importance of wildlife.

[2]

- (b)** Describe sustainable ways of dealing with problem animals in Zimbabwe.

[3]

(c) Describe the following terms used in Agriculture:

(i) biodiversity,

_____ [1]

(ii) poaching,

_____ [2]

(iii) ecology.

_____ [2]

2 (a) Describe the application of lime in seedbed preparation.

_____ [4]

- (b) An experiment was carried out to determine the amount of air in a given soil sample. **Fifty millimetres** of water was added to **50 ml** of an oven dry soil sample. After **30** minutes the reading in the cylinder with a mixture of water and soil was **80 ml**.

Calculate the percentage of air in the soil sample.

[4]

- (c) Explain any **one** cultural way of preventing soil erosion in an arable land.

[2]

- 3 (a) Explain the importance of crop breeding to the farmer.

[4]

- (b) Differentiate between open pollinated varieties and hybrids.

[2]

- (c) Describe the term *heterosis*.

[2]

- (d) Describe the following terms:

- (i) emergency,

[1]

- (ii) germination.

[1]

- 4 (a) Identify any **five** socio-economic importance of livestock to the farmer.

[5]

- (b) Discuss any **three** reasons for dehorning livestock.

[3]

- (c) Give the functions of the following female reproductive parts:

- (i) vagina,

[1]

- (ii) uterus.

[1]

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5 (a) Explain the functions of the listed maize sheller parts:

(i) roller,

_____ [1]

(ii) base plate,

_____ [1]

(iii) pressing fan,

_____ [1]

(iv) threshing cylinder,

_____ [1]

(v) feed tray.

_____ [1]

(b) Explain the operational principle of a traditional grain storage facility.

 _____ [2]

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8

- (c) Calculate, rolls of barbed wire required for a **four** strand fence, on a plot measuring **200 m × 80 m**. Standard roll of barbed wire is **700 m**.

[3]

- 6 (a) Explain any **three** importance of financial records to a farmer.

[3]

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- (b) A farmer harvested **6 tonnes** of maize and sold it at **\$300** per tonne.
The following payments were made:

Compound D	- \$200,00,
Chemicals	- \$ 50,00,
Farm manager's salary	- \$400,00,
Urea	- \$150,00,
Farm licence	- \$100,00,

Calculate the gross margin for the enterprise. (Show all working)

[3]

- (c) Identify any **two** possible causes of change in prices of agricultural products.

[2]

- (d) Give a solution to each of the following risks:

- (i) theft,

[1]

- (ii) fire.

[1]

10
SECTION B [20 marks]

Answer **one** question only from this section.

CROP HUSBANDARY

- | | | |
|----------|---|-----|
| 7 | (a) Outline the socio-ecological importance of forestry. | [8] |
| | (b) Discuss the process of pricking out in gumtree establishment. | [4] |
| | (c) Explain the management practices carried out in an indigenous tree plantation. | [8] |
| 8 | (a) Describe toxicity levels of agro-chemicals. | [8] |
| | (b) Explain advantages of traditional tillage practices. | [7] |
| | (c) Design, in sequence, a four year rotation for a field crop. | [5] |
| 9 | (a) Outline the advantage and disadvantages of organic fertiliser in crop production. | [9] |
| | (b) Evaluate the use of traditional grain storage facilities. | [5] |
| | (c) Describe ways of reducing the effects of environmental factors on crop production. | [6] |

11
SECTION C [20 marks]

Answer **one** question only from this section.

- 10** **(a)** Discuss characteristics of an acceptable fowl run. [5]
- (b)** Outline functions of any **four** parts of the digestive system of a **named** four legged non-ruminant animal. [8]
- (c)** Describe desirable characteristics of a **named** breeding female of a non-ruminant animal. [7]
- 11** **(a)** State and explain **two** types of livestock immunity. [4]
- (b)** Describe the Animal Health Act of Zimbabwe. [7]
- (c)** Outline management operations carried out on a calf from birth to weaning. [9]
- 12** **(a)** Describe the management of indigenous chicks from **day old** to **two weeks** of age. [8]
- (b)** Describe the life cycle of a **named** internal parasite of a farm animal. [7]
- (c)** Evaluate the importance of roughage in the diet of ruminant farm animals. [5]